27. Wednesday, June 4. **Finite and infinite sets** (9.1)

Definition 1: Let A and B be sets. We say that A and B are **equinumerous** (called equivalent in the textbook) if there exists a bijection $A \longrightarrow B$.

Notation: $A \approx B$ → \approx

Fact: If A and B are finite, then

$A \approx B \iff A$ and B have the same number of elements.

Note: If A and B are infinite, funny things can happen

E.g. $\mathbb{N} \approx \mathbb{Z}$, $(0, \infty) \approx \mathbb{R}$, $(-\pi/2, \pi/2) \approx \mathbb{R}$.

(We gave bijections between these sets)

Stuff about bijections:

1) Define $I_A : A \to A$ by $I_A(x) = x \ \forall x \in A$. The identity function on A

Then I_A is a bijection.

$$A \approx A$$

2) If $f : A \to B$ is bijective, Then $f^{-1} : B \to A$ is bijective.

$$A \approx B \implies B \approx A$$

3) If $f : A \to B$ and $g : B \to C$ are bijective, Then $g \circ f : A \to C$ is bijective

$$A \approx B \text{ and } B \approx C \implies A \approx C.$$

These facts prove The following:

Theorem 2: The relation $\approx$ is an equivalence relation (on The collection of all sets

Finite sets (9.1)

Recall: When we studied finite sets before, for a finite set A we defined the cardinality of A as

$$\text{card}(A) = |A| = \text{The number of elements in } A.$$

Now we can refine this.

Definition 3: For each $k \in \mathbb{N}$, define $\mathbb{N}_k = \{1, 2, \ldots, k\}$ Also denoted $[k]$

Definition 4: A set A is called **finite** if either $A = \emptyset$ or there exists some $k \in \mathbb{N}$ such that $A \approx \mathbb{N}_k$.

We say that A is **infinite** if it's not finite.

Fact: For any $j, k \in \mathbb{N}$, if $\mathbb{N}_j \approx \mathbb{N}_k$ then $j = k$. So it makes sense to define:

Definition 4 (continued): If $A \approx \mathbb{N}_k$, we say that A has **cardinality** k and write $\text{card } A = k$.

The empty set has cardinality 0, so $\text{card } \emptyset = 0$.

Theorem 5: If A is a finite set with card $A = k > 0$ and $B \approx A$, Then card $B = k$.

Proof: Suppose A is finite with card $A = k$. Then, by definition, $A \approx \mathbb{N}_k$.

Since $B \approx A$, by Transitivity, $B \approx \mathbb{N}_k$. Hence card $B = k$. □

Lemma 6: If A is a finite set and $x \notin A$, Then $A \cup \{x\}$ is a finite set and

$$\text{card}(A \cup \{x\}) = \text{card}(A) + 1.$$

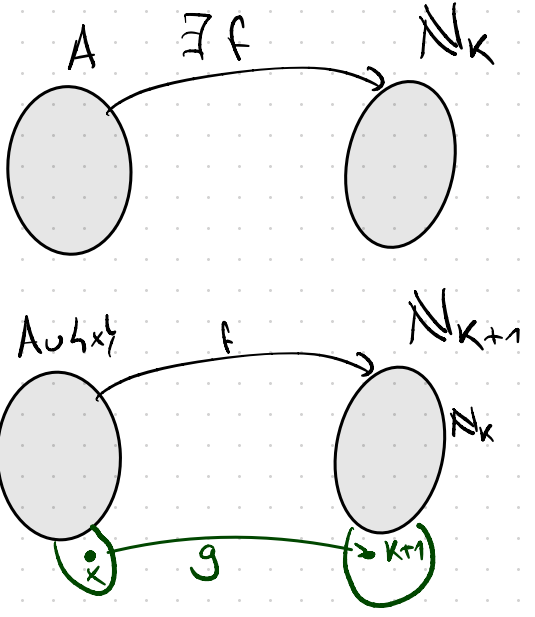
Proof sketch: Let $k = \text{card } A$, so $A \approx \mathbb{N}_k$.

Hence $\exists$ bijection $f: A \longrightarrow \mathbb{N}_k$.

Now extend it to $g: A \cup \{x\} \longrightarrow \mathbb{N}_{k+1}$,

$$g(a) = f(a) \text{ if } a \in A$$
$$g(x) = k+1$$

Prove g bijective. □



Corollary 7: If A is a finite set and $x \in A$, Then $A \setminus \{x\}$ is finite and
$\text{card}(A \setminus \{x\}) = \text{card}(A) - 1$

Proof sketch: Use $A = (A \setminus \{x\}) \cup \{x\}$ and lemma 6 □

with ~~x~~

Lemma 8: Let $k \in \mathbb{N}$. If $A \leq \mathbb{N}_k$, Then A is finite and $\text{card}(A) \leq k$ — $P(k)$

Proof sketch: Induction on k. Assume $P(k)$ true.

For $P(k+1)$, Let $A \leq \mathbb{N}_{k+1}$.

- If $k+1 \notin A$ Then $A \leq \mathbb{N}_k \Rightarrow \text{card } A \overset{IH}{\leq} k < k+1$ ✓
- If $k+1 \in A$, then $A \setminus \{k+1\} \leq \mathbb{N}_k \Rightarrow \text{card}(A \setminus \{k+1\}) \overset{IH}{\leq} k$
  $= \text{cor } 7$
  $\text{card}(A) - 1$

□

All these results now allow us to prove:

Theorem 9: If S is a finite set and $A \subseteq S$, Then A is finite and

$$\text{card } A \leq \text{card } S$$

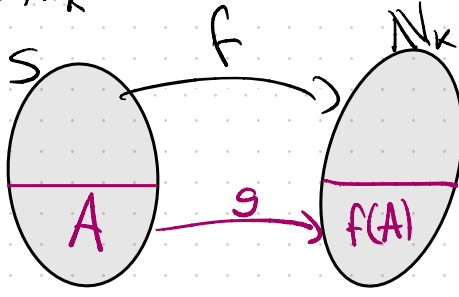
Proof sketch: Let $k = \text{card } S$, so $\exists$ bijection $f: S \longrightarrow \mathbb{N}_k$

Step 1: "Restrict f to A", get function $g: A \longrightarrow f(A)$,
$g(x) = f(x)$ for $x \in A$

Step 2: prove g bijective.

Step 3: $\text{Card } A = \text{card } f(A) \leq k$
(lemma 8)

$g: A \longrightarrow f(A)$ bijective.



Corollary 10: Suppose S is a finite set and there exists an injection $A \longrightarrow S$ Then A is finite and $\text{card } A \leq \text{card } S$.

Proof Let $f: A \longrightarrow S$ injective. Consider the subset $f(A) \subseteq S$

Define $g: A \longrightarrow f(A)$ by $g(x) = f(x)$ for $x \in A$.

Claim 1: g is injective: In fact.

If $x, y \in A$ have $g(x) = g(y) \Rightarrow f(x) = f(y) \Rightarrow x = y$, because f injective.

Claim 2: g surjective: In fact,

given $y \in f(A)$, by definition of $f(A)$ there exists $x \in A$ such that $f(x) = y$

But Then $g(x) = f(x) = y$, so $g(x) = y$, which proves claim 2.

Hence, by Claims 1 and 2, $g: A \longrightarrow f(A)$ is bijective, so $A \approx f(A)$

Therefore $\operatorname{card} A = \operatorname{card} f(A) \leq \operatorname{card} S$

g bijective — Thm 9. □

The contrapositive of corollary 10 gives us for free:

Theorem (The pigeonhole Principle).

If A and B are finite sets with $\text{card}(A) > \text{card}(B)$, Then there is no injection $A \longrightarrow B$

Note: If m pigeons are sitting on r pigeonholes where $m > r$, Then at least one pigeonhole contains more Than one pigeon.

Proof: Let $P = \{\text{pigeons}\}$, so $\text{card}(P) = m$

Let $H = \{\text{pigeonholes}\}$ so $\text{card}(H) = r$

"Sitting" means we have a function

$f : P \longrightarrow H$ where $f(\text{pigeon}) = \text{pigeonhole where it sits.}$

f cannot be injective, so Two pigeons sit in The same pigeonhole. □